

Stochastic Process

$\Rightarrow$ Collection of random variables indexed by time

$x_0, x_1, x_2, x_3 \}$ Discrete

$\{x_t\}_{t \geq 0}$ Cont.



$\Rightarrow$ Probability distribution over a space of paths.

Examples $\rightarrow$

① $f(x) = x$ prob. $\rightarrow 1$



② $f(x) = x \forall x$ prob $\rightarrow 1/2$
 $f(x) = -x \forall x$ prob $\rightarrow 1/2$

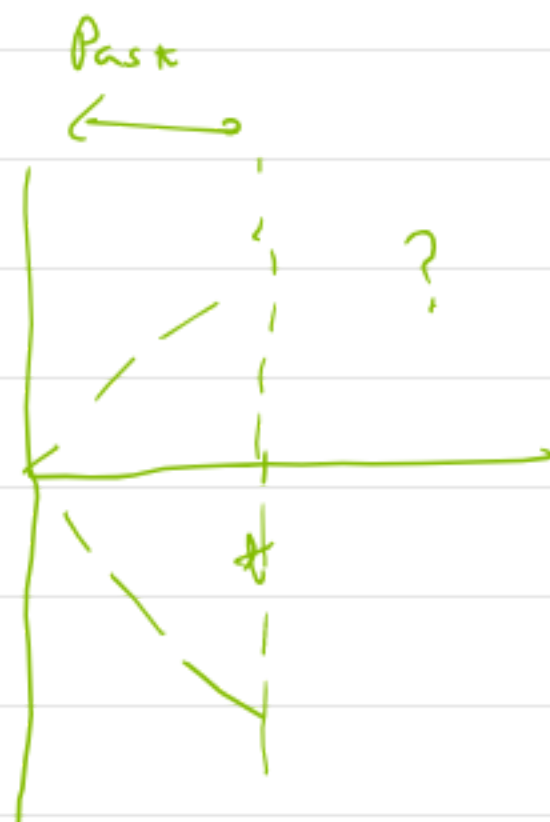


Interdependent

③ for each x $f(x) = x$ ($1/2$)
 $= -x$ ($1/2$)



Example



a) What are the dependencies?

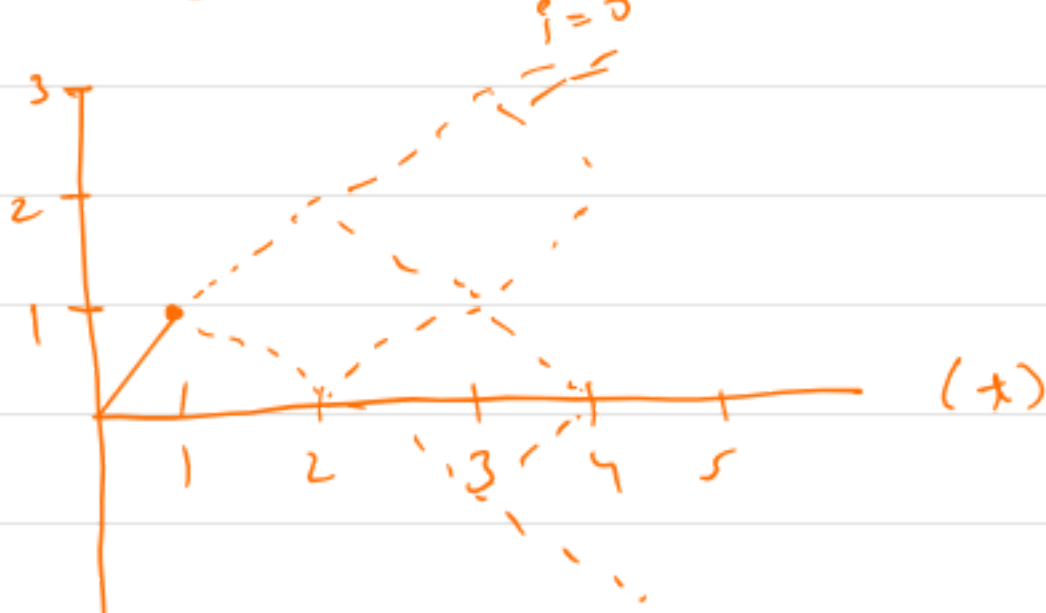
b) What is long term behaviour?

c) What are the boundary events?
(extreme behaviour)

Simple Random Walk (Discrete)

Def Y_i : iid random variables $\begin{cases} 1 & \text{prob}(1/2) \\ -1 & \text{prob}(1/2) \end{cases}$

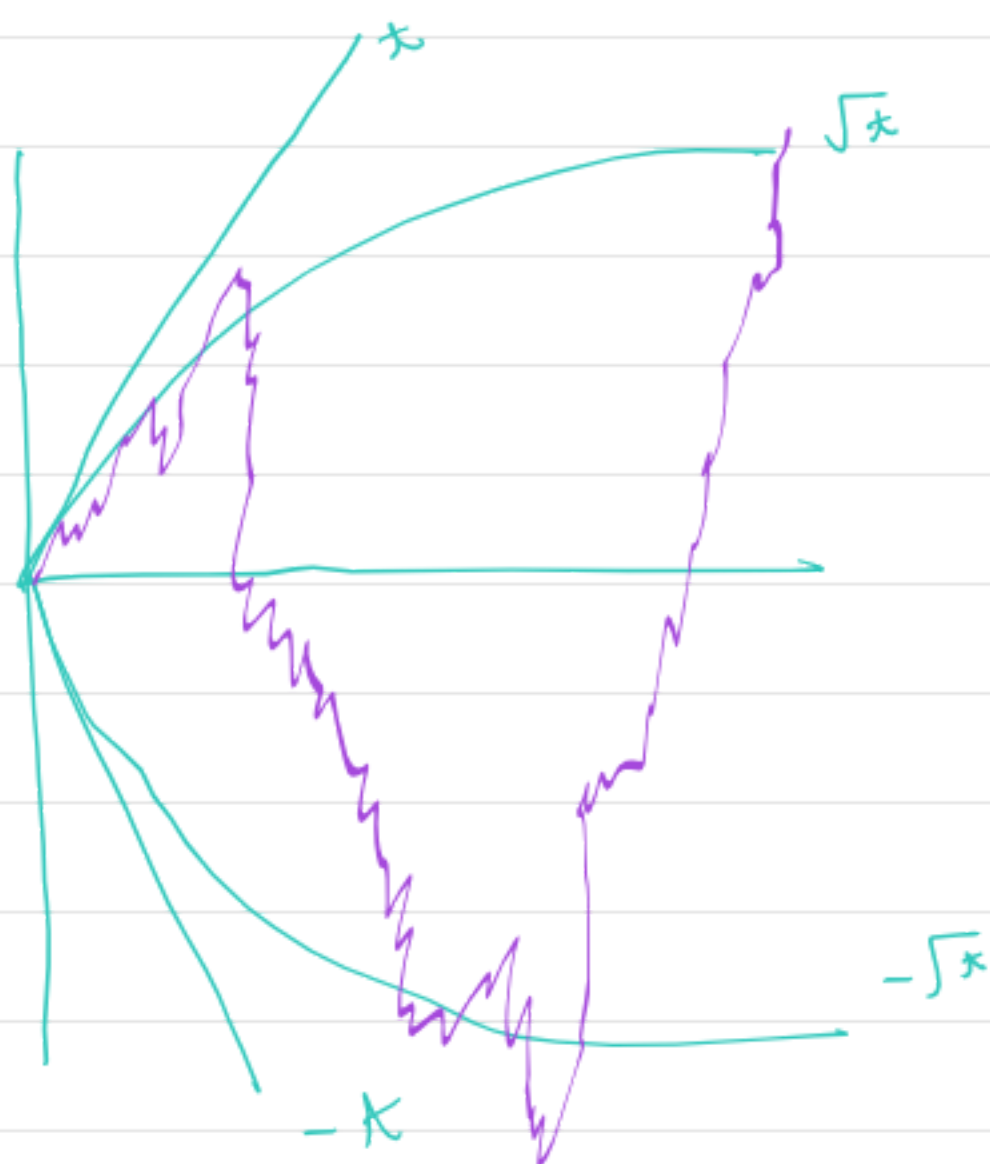
For each t , $X_t = \sum_{i=0}^t Y_i$, $X_0 = 0$



$\Rightarrow$ for very large t ,

variance $\rightarrow t$

std. deviation $\rightarrow \sqrt{t}$



Properties $\rightarrow$

1) $E(X_k) = 0$ $\left(\left(+\frac{1}{2} \right) + \left(-\frac{1}{2} \right) \right)$

$\Rightarrow$ No preferred direction

2) Independent movement.

3) Stationary distribution,

For $h \geq 1$, $X_{t+h} - X_t$ has same dis.
as X_h

$\Rightarrow$ Not stationary $S_{t+h} - S_t \neq S_h$

Examples

1) Coin Toss game

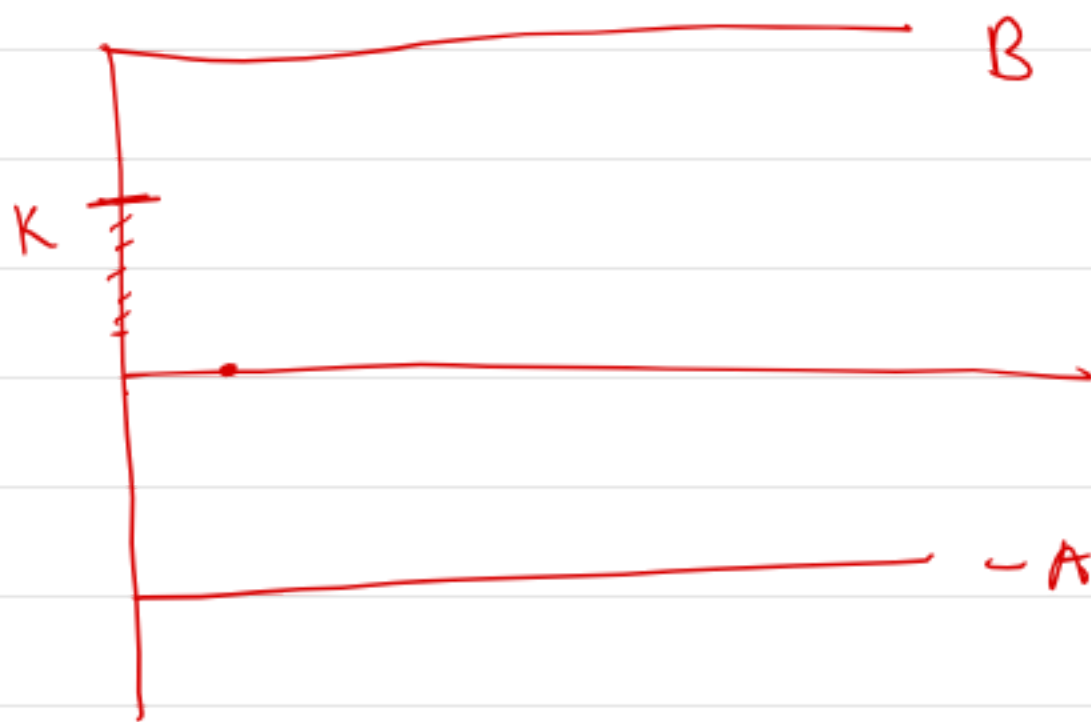
Alice $\rightarrow$ Heads
Wins 1\$

Bob $\rightarrow$ Tails
Wins 1\$

Alice will stop at +100\$ or -100\$
 $\uparrow$ Prob

$\Rightarrow$ By symmetry, Alice will get +100 $\frac{1}{2}$ times

$\Rightarrow$ Now +100\$ or -50\$



$P(k) \Rightarrow P(\text{hit line B first before A})$

$$P(-A) = 0 \quad P(B) = 1$$

$$P(k) = \frac{1}{2} P(k+1) + \frac{1}{2} P(k-1) \quad \Leftarrow \text{Two ways, } \frac{1}{2} \text{ up, } \frac{1}{2} \text{ down}$$

How to Solve Recursion

$$\rightarrow f(x) = f(x-1) + f(x-2)$$

$$f(0) = 1 \quad f(1) = 1$$

Fibonacci Series ϕ

$$x^2 = x + 1$$

α, β

$$\frac{1+\sqrt{5}}{2} \quad \frac{1-\sqrt{5}}{2}$$

$$f_n = c_1 (\alpha)^n + c_2 (\beta)$$

$$f_n = c_1 \left(\frac{1+\sqrt{5}}{2} \right) + c_2 \left(\frac{1-\sqrt{5}}{2} \right)$$

dp $\rightarrow O(n)$
formula $\rightarrow O(1)$

$$1s = 10^8$$

3^{900}

$$ans = 1$$

$$3^{55}$$

$$55 = 110111$$

$$3^{55} = 3^{32} \times 3^{16} \times 3^7 \times 3^2 \times 3^1$$

$$10^{18} \approx 2^{67}$$

$$10^9 \approx 2^{32}$$

$$10^6 \approx 2^{16}$$

$$3^1 \rightarrow 3^2 \rightarrow 3^4 \rightarrow 3^8 \rightarrow 3^{16} \rightarrow 3^{32}$$

$$f(x) = a f(x-1) + b f(x-2)$$

$$T_0, T_1, T_2, T_3, T_4$$

$$\begin{bmatrix} T_2 \\ T_1 \end{bmatrix} = \begin{bmatrix} a & b \\ 1 & 0 \end{bmatrix} \begin{bmatrix} T_1 \\ T_0 \end{bmatrix}$$

$\uparrow \quad \uparrow$
 $P \quad A$

$$\begin{bmatrix} T_3 \\ T_2 \end{bmatrix} = \begin{matrix} \text{Transformation Matrix} \\ \begin{bmatrix} a & b \\ 1 & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} a & b \\ 1 & 0 \end{bmatrix} \begin{bmatrix} T_1 \\ T_0 \end{bmatrix}$$

$$\begin{bmatrix} T_n \\ T_{n-1} \end{bmatrix} = \begin{matrix} \downarrow^{n-1} \\ A \end{matrix} \begin{bmatrix} T_1 \\ T_0 \end{bmatrix}$$

$$f(x) = a f(x-1) + b f(x-2) + c f(x-3)$$

$$\begin{bmatrix} T_n \\ T_{n-1} \\ T_{n-2} \end{bmatrix} = \begin{bmatrix} a & b & c \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} T_2 \\ T_1 \\ T_0 \end{bmatrix}$$

$$(3) O(\log_2(n))$$

Matrix

$$O(n)$$

dp

$$O(\log_2 n)$$

Formula

Markov Chain

Stochastic Processes whose effect of the past on the futuneness summarized only by the current state.

=> Let's say discrete ->

$f(k) \leftrightarrow f(k-1)$ dependent

$f(k+1) \leftrightarrow f(k-1)$ independent



Ex. -> Simple Random Walk

SRW

If X_t have values in S (Finite Set)

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

$$\sum_j P_{ij} = 1 \quad (\text{Will go somewhere})$$

1 -> 2 1 -> 1 1 -> 3
2 -> 2 1 -> 2 3 -> 2

$$P_{12} P_{22} + P_{11} P_{12} + P_{13} P_{32}$$

Transition
Prob.
Matrix

=

P_{11}
 P_{12}
 $\vdots$
 P_{1n}

P_{12} (1 step)

(2 steps) P_{21} - - - P_{n1}

P_{ij}

P_{mn}

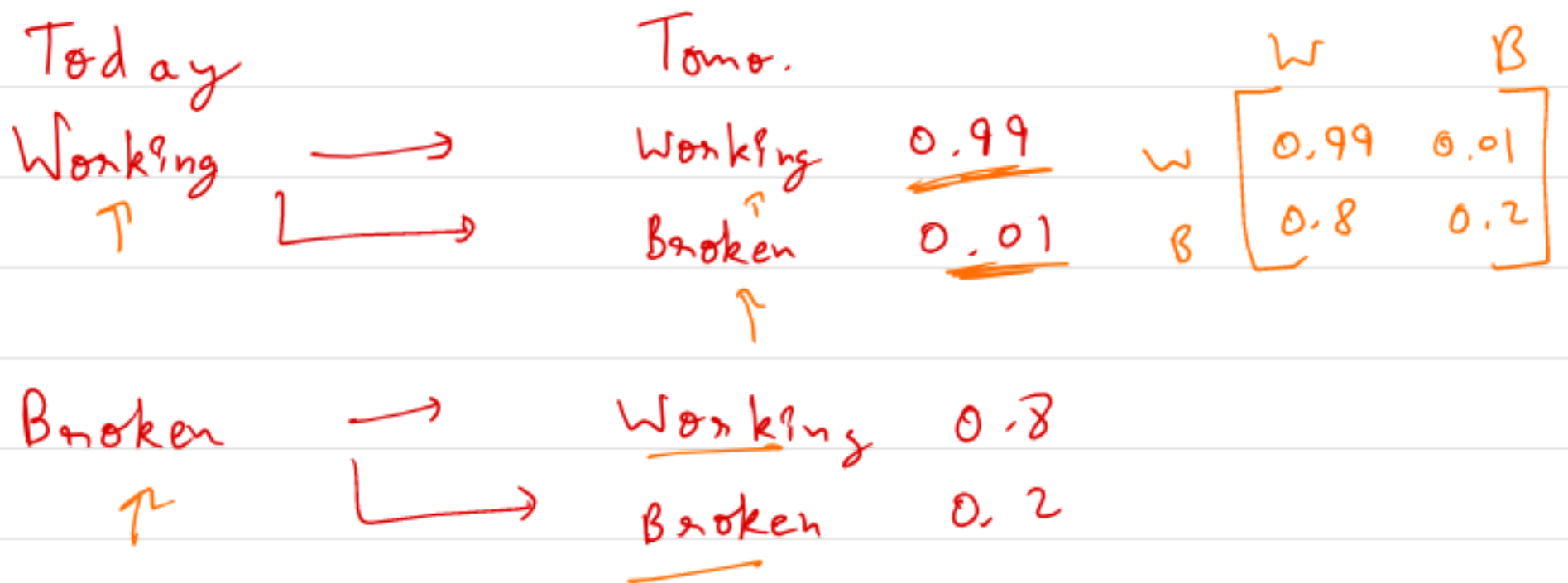
P_{11}	P_{12}	P_{13}
P_{21}	P_{22}	P_{23}
P_{31}	P_{32}	P_{33}

Not
for SRW,
As S is
infinite

=> Increase steps => $A^2, A^3, A^4, \dots$



Ex. Machine broken or Working at a given day



Two States { Working, Broken }

W B

$$W \begin{bmatrix} 0.99 & 0.1 \\ 0.8 & 0.2 \end{bmatrix} B = A \text{ (Transition Prob. Matrix)}$$

$\Rightarrow$ After 1 year (365 days)

$$\begin{bmatrix} A \end{bmatrix}^{365} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

$\downarrow$

1 $\rightarrow$ Working

2 $\rightarrow$ Broken

$\Rightarrow$ Over long Period, $\underline{A^{3651}} \approx \underline{A^{3150}}$

$$A^{3651} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = A^{3650} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

A $\rightarrow$ tends to Eigen values $\rightarrow \begin{bmatrix} p \\ q \end{bmatrix}$

$$A \begin{bmatrix} p \\ v \end{bmatrix} = \begin{bmatrix} p \\ v \end{bmatrix}$$

Perron-Frobenius (what's on the board) just says: if all entries of A are positive, there's guaranteed to be one **dominant positive eigenvalue** with a positive eigenvector. Used in Markov chains to find long-run steady states.

$$A \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\boxed{\lambda = 1} \quad (\text{By solving})$$

↪ In general answer

⇒ If transition matrix A has +ve entries, then $\exists$ direction $v \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_n \end{bmatrix}$ s.t. $Av = v$

⇒ Such vector v is called stationary distribution.

⇒ Consider Prob. Distribution X over S distribution. such that $P(X=i) = \pi_i$

What this means:

You have a set of states S (e.g. $S = \{1, 2, 3\}$).

X is a random variable that takes values from S with some probabilities.

π_i is just the probability that X lands on state i .

So if $S = \{1, 2, 3\}$:

- $P(X=1) = \pi_1 = 0.5$
- $P(X=2) = \pi_2 = 0.3$
- $P(X=3) = \pi_3 = 0.2$

All π_i must sum to 1 (since total probability = 1).

Martingale

=> Fair Game

$$X_t = \overset{\text{expectation}}{E} [X_{t+1} | S] \quad \forall t \geq 0$$

$$S = \{x_0, x_1, x_2, x_3\}$$

$E_x \rightarrow$ SRW

=> Each step has same outcome as prev.

2. $Y_1, Y_2, \dots$ iid random

$$Y_i = \begin{cases} 2 & (\text{prob } 1/3) \\ 1/2 & (\text{prob } 2/3) \end{cases}$$

$$X_0 = 1 \quad X_k = \prod_{i=1}^k Y_i$$

Outcome doesn't change by history

$$\text{Expectation} = \boxed{1}$$

Optional Stopping Theorem

Def Stopping Time $\rightarrow$

Given a stochastic process $\{X_0, \dots, X_t\}$
a non-negative integer τ is called a
stopping time if integer $k \geq 0$, $\tau \leq k$ depends on
 $X_0, \dots, X_k$.

Discrete
 $\downarrow$ (valid for both)

Ex. In coin toss game, stop when balance = \$100
 $\tau \rightarrow$ stopping time

or
-\$50

$k = t$
 $\downarrow$
 $k = t + 1$

i) Stop on first peak $\rightarrow$ Depends on $t+1$
Not stopping time.

$\Rightarrow$ Theorem

Suppose $X_0, X_1, \dots$ is a martingale, τ is a
stopping time, furthermore $\exists$ constant T s.t.
 $\tau \leq T$ always.

$$\text{Then } E[X_\tau] = X_0$$

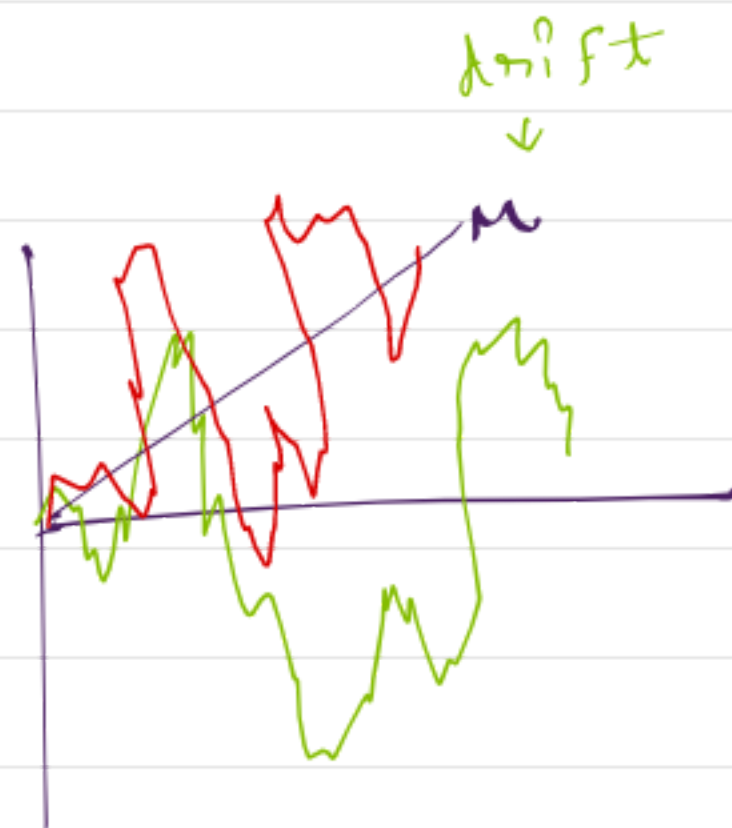
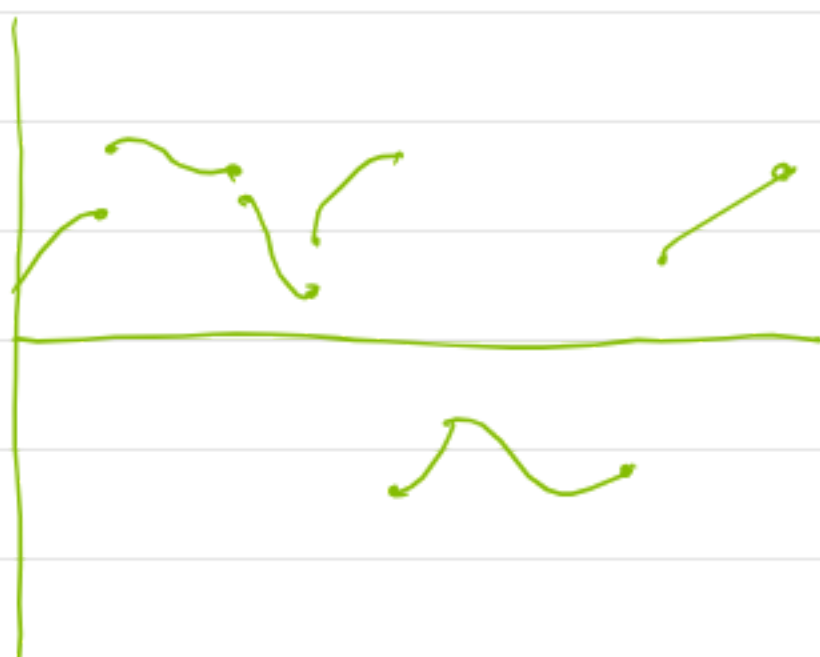
Case 1

$$k(100) + (1-k)(-50) \Rightarrow 150k - 50 = 0$$

$k = 1/3$

Continuous Stochastic Processes

$\{x_t\}_{t \geq 0}$ not Integer



- How to describe prob. distribution?
 - ↳ prob. density function

Brownian Motion

Probability distribution over the set of
cont. functions $B: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$

1) $P(B(0) = 0) = 1$ Starts at zero. Always

2) Stationary increment $N(\text{mean}, \text{variance})$
 $B(t) - B(s) \sim N(0, t-s)$ $\propto \sqrt{t}$

⇒ The change in B over interval $[s, t]$

- Mean = 0, No drift, (↑, ↓) same prob.

⇒ Variance = $t-s$, uncertainty grows with t .

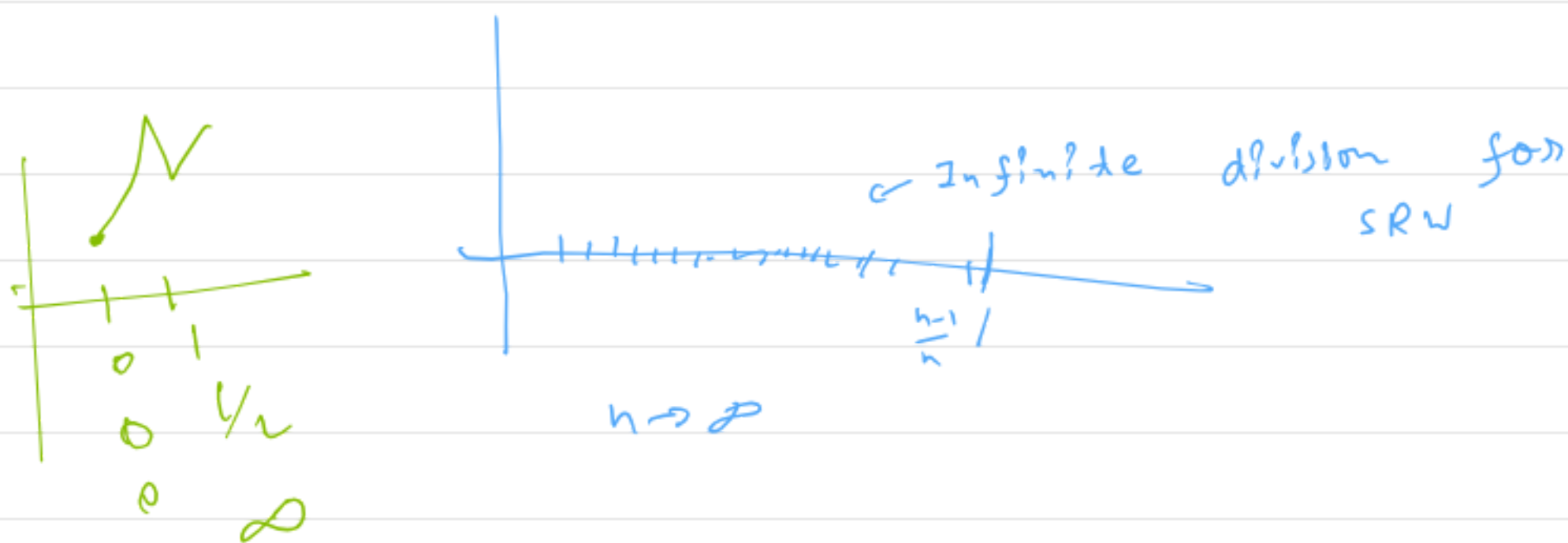
3) Independent increments,
Two non overlapping intervals $\rightarrow$ independent.

This Prob. dist. is Brownian Motion.
(Weiner Process)

$\Rightarrow$ We are finding prob. for a certain path $\star\star$.

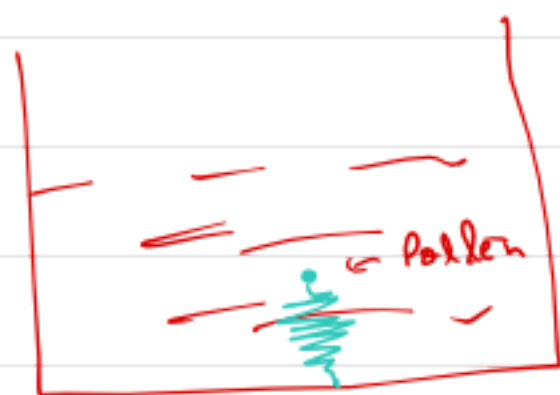
$\Rightarrow$ Brownian Motion is limit of Simple Random Walk.

Divide discrete values to small parts



Ex. Botanist Robert Brown

$\Rightarrow$



Random path in 2d
while going down

Einstein $\rightarrow$ explained phy
Water molecule collides
with pollen (affect accumulates)

Ex 2. Stock Prices [★]

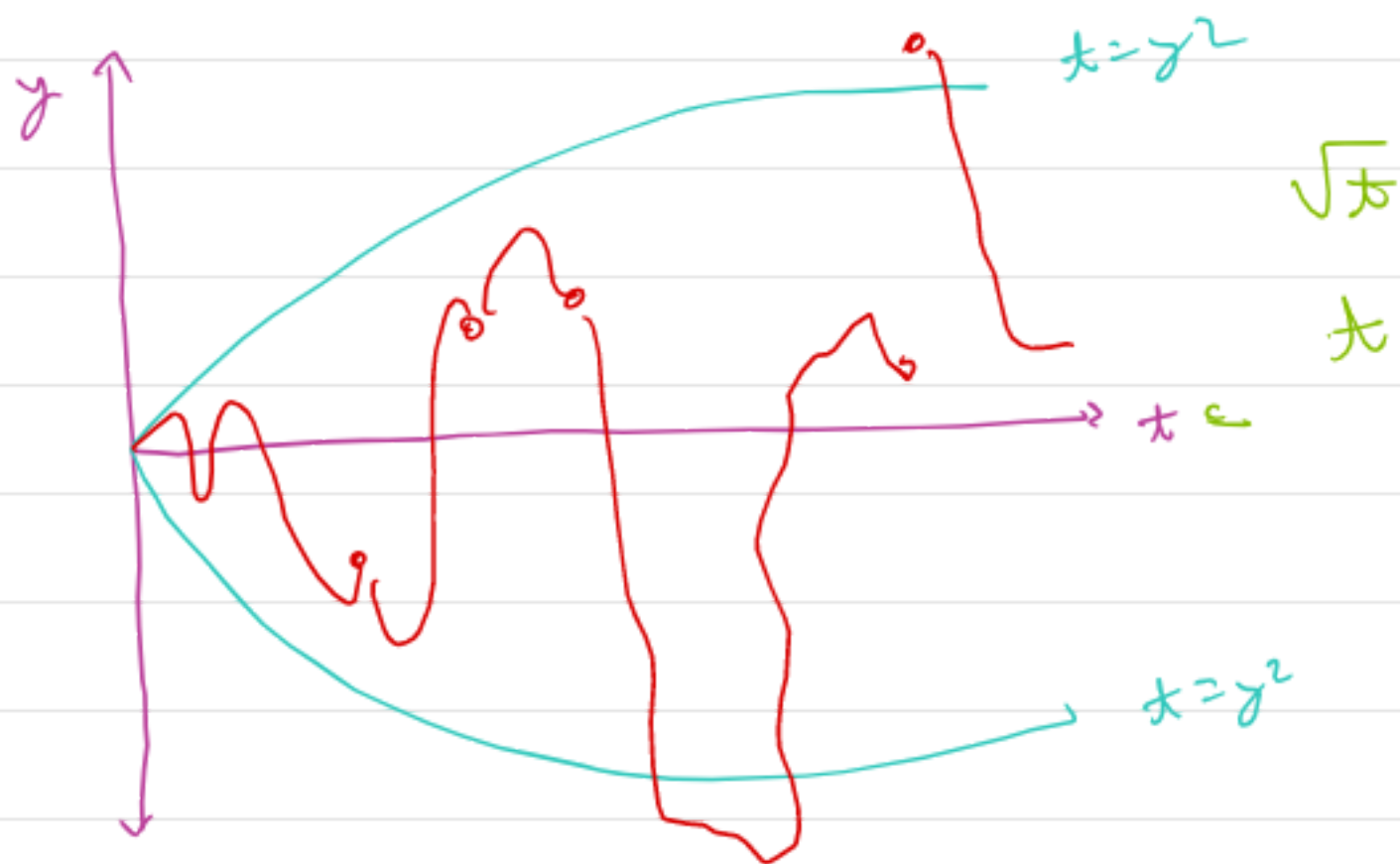
Buy & Sell $\rightarrow$ Moves up and down

Motion $\approx$ Brownian Motion

Properties $\rightarrow$

1) Crosses t axis infinitely often.

2) Does not deviate much from $t = y^2$



3) Not differentiable anywhere

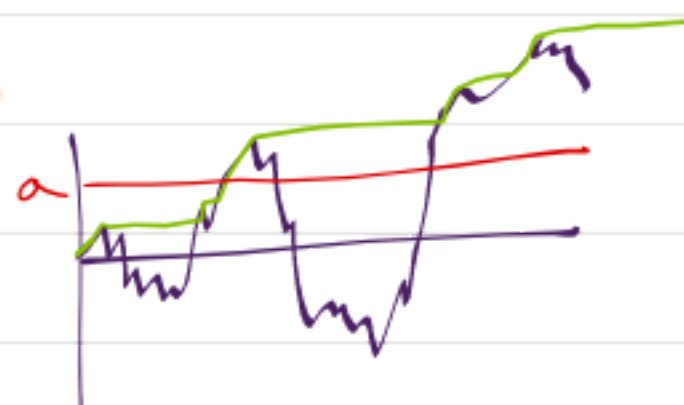
Can't use calculus

* Using Brownian motion as model



for daily
stock price

$$M(t) = \max_{s: s \geq t} (B(s))$$



Theorem

For all $t > 0$ and $a > 0$

$$P(M(t) > a) = 2 P(B(t) > a)$$

Proof

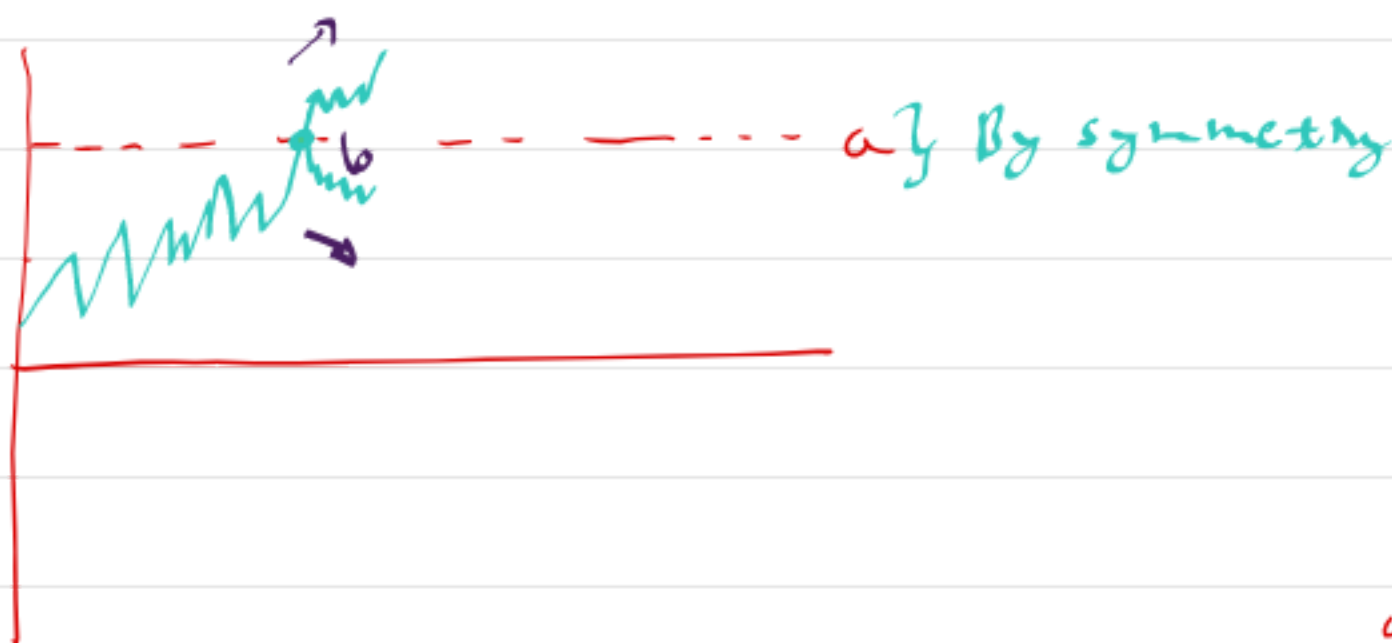
$$\tau_a = \min \{ B(t) = a \}$$



$\Rightarrow$ Time when it crosses $B(t) = a$ line first time.

$$B(t) > B(\tau_a)$$

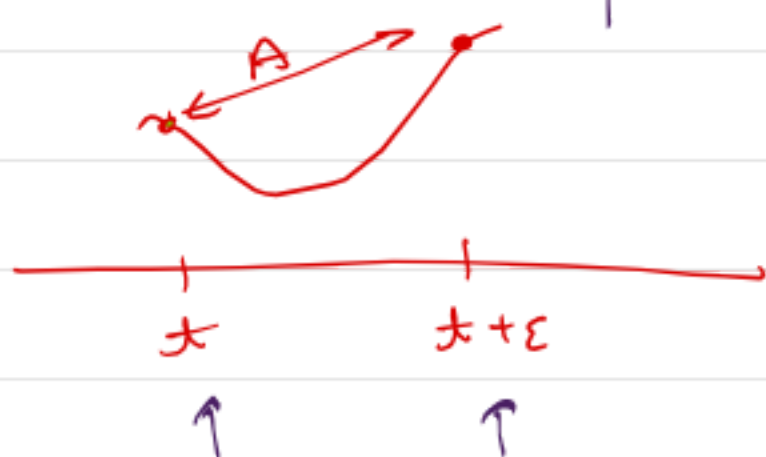
$$P(B(t) - B(\tau_a) > 0 \mid \tau_a < t) = \frac{P(B(\tau_a) > B(t))}{P(B(t) - B(\tau_a) < 0 \mid \tau_a < t)}$$



$$P(M(t) > a) = P(z_u < -a) = 2 P(B(t) - B(z) < 0 \mid z_a < t) \\ = 2 P(B(t) > a)$$

Th For each $t \geq 0$, the Brownian motion is not differentiable at t with prob. 1

Proof



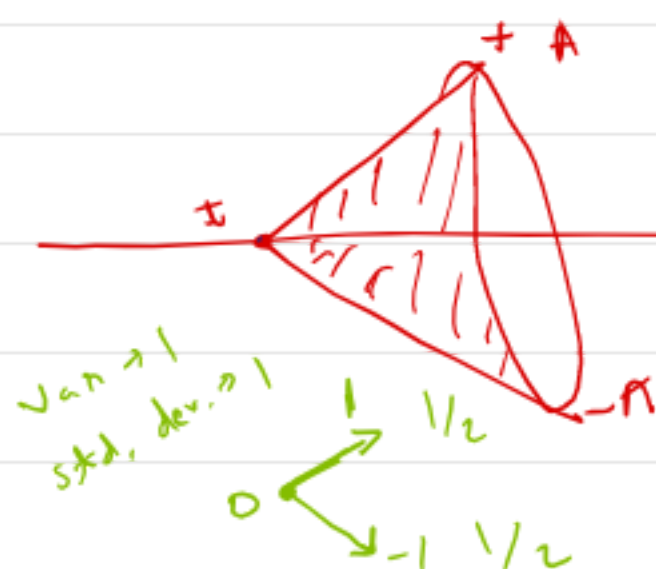
Suppose $\frac{dB(t)}{dt} = A$

then

$$|B(t + \epsilon') - B(t)| \leq \epsilon A$$

$$\dots \dots \dots \forall \epsilon' \leq \epsilon$$

$$M(\epsilon) \leq \epsilon A$$



value remain inside this cone

$$P(M(\epsilon) > \epsilon A) \\ = 2 P(B(\epsilon) > \epsilon A) \\ = 2 P(N(0, \epsilon) > \epsilon A)$$

So if $M(\epsilon)$ is outside cone $\Rightarrow$ Statement $\times$

$$P(M(\epsilon) > \epsilon A) = 2 P(B(\epsilon) > \epsilon A) \\ = 2 P(N(0, \epsilon) > \epsilon A) \quad \text{Divide by } (\sqrt{\epsilon}) \\ = 2 P(N(0, 1) > \sqrt{\epsilon} A)$$

$$2 P(N(0, 1) > \sqrt{\epsilon} A) \\ = 2 P(N(0, 1) > 0) \\ = \frac{1}{2} \times 2 \\ = 1$$

As $\epsilon \rightarrow 0$ $\sqrt{\epsilon} A \rightarrow 0$

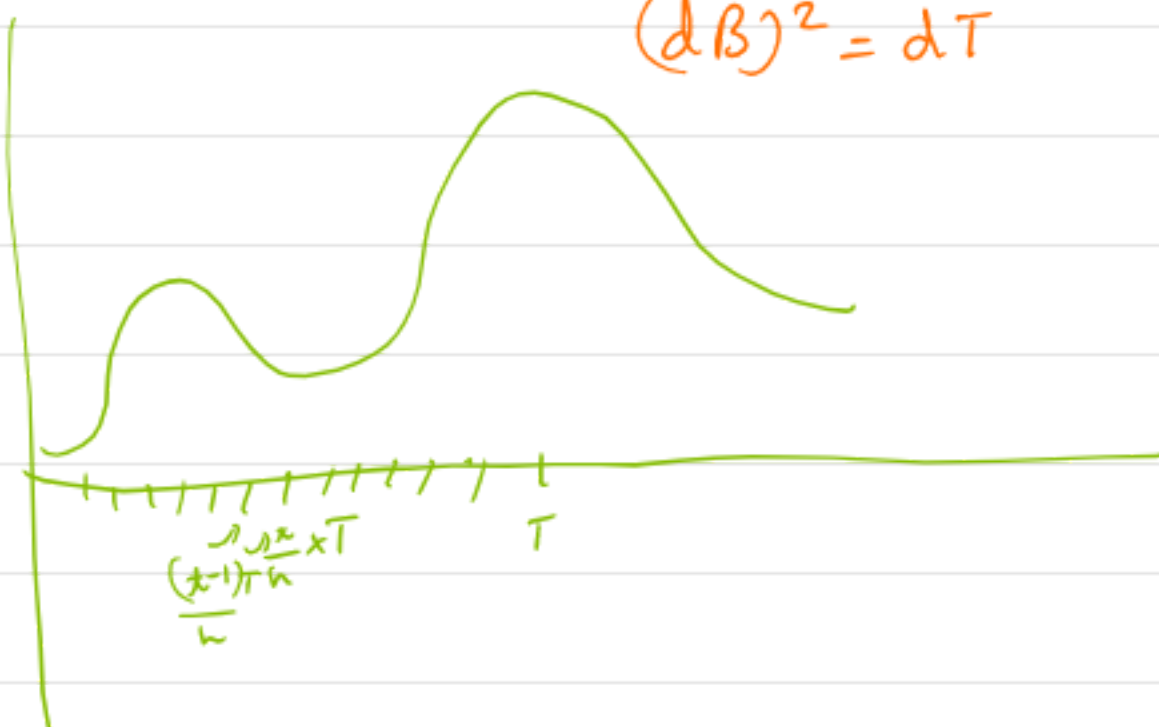
$$P(\cdot) \rightarrow \frac{1}{2} \quad P(M(\epsilon) > \epsilon A) = 1$$

Theorem $\rightarrow$ Quadratic Variation

$\forall T > 0$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(B\left(\frac{t_i}{n} \times T\right) - B\left(\frac{t_{i-1}}{n} \times T\right) \right)^2 = T$$

$$(dB)^2 = dT$$



Assume f is continuously differentiable,
Then,

$$\sum_{i=1}^n \left(f(t_i) - f(t_{i-1}) \right)^2 \leq f'(s_i) (t_{i+1} - t_i)^2$$

$$s_i \in [t_i, t_{i+1}]$$

$$= \max_{0 \leq s \leq T} \left(f'(s) \right)^2 \sum_{i=1}^n (t_{i+1} - t_i)^2$$

$$= \frac{T^2}{n} \rightarrow 0$$

$\Rightarrow$ Variation too much in very small diff.,
So not differentiable.

$$\sum_{i=0}^{n-1} \left(B(t_{i+1}) - B(t_i) \right)^2 \quad t_i = \frac{i}{n} T$$

$$N \left(0, \frac{T}{n} \right)$$

$$N \left(0, \frac{T}{n} \right) = X_i$$

$$\sum X_i^2$$

Mean variance = $t_{i+1} - t_i$

$$\sum_{i=0}^{n-1} X_i^2$$

$$X_i \sim N \left(0, \frac{T}{n} \right)$$

$$Y_i = X_i^2 \frac{n}{T}$$

$$E X_i^2 = \frac{T}{n}$$

$$E Y_i = \frac{T}{n}$$

} Variance

$$\frac{T}{n} \left(\sum_{i=0}^{n-1} Y_i \right)$$

=> We divide by variance, so it standardizes to 1

=> In Stock Prices, we want the %ile diff. to be normally distributed

$$\frac{ds_t}{s_t} = dB_t \quad \text{Brownian}$$

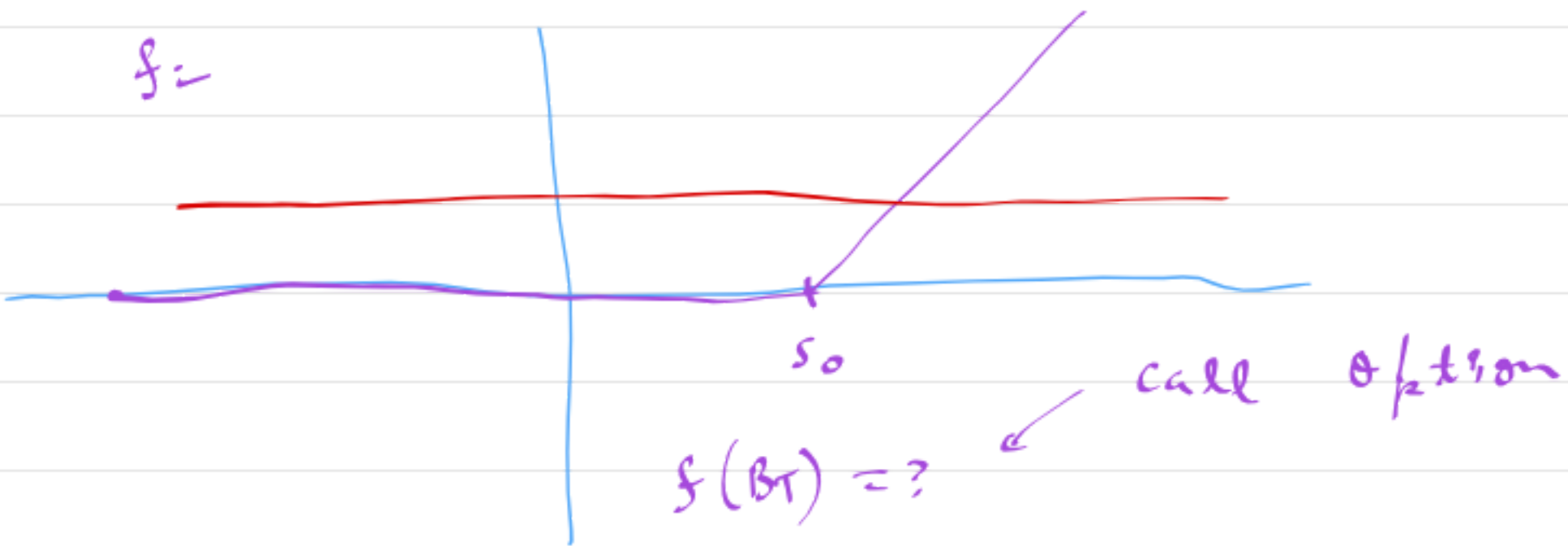
$$s_t = e^{B_t}$$

X

Ito Calculus *

Suppose we want to compute $f(B_t)$,
for some smooth function f ,

Estimate Infinitesimal difference.



If B_t was diff.

$$df = \left(\frac{dB_t}{dt} \right) dt \quad \times$$

$$\Rightarrow df = f'(B_t) dB_t \quad \leftarrow \text{wrong}$$

↳ computing small change

because $(dB)^2 = dt$

$$f(t+x) = f(t) + \frac{f'(t)x}{1} + \frac{f''(t)x^2}{2} - \dots - \dots$$

$$f(t+x) - f(t) = f'(t)x + \frac{f''(t)x^2}{2} - \dots$$

In classical world

$$\frac{f(t+x) - f(t)}{x} = f'(t)$$

$$f(t+x) - f(t) \propto f'(t)x$$

$$\Rightarrow f(B_{t+\tau}) - f(B_t) = f'(B_t) (B_{t+\tau} - B_t) + \frac{f''(B_t)}{2} (dB_t)^2$$

Not ignorable

$\Rightarrow$ So finally,

$$df = f'(B_t) dB_t + \frac{f''(B_t)}{2} dt$$

small changes
in f

Ito's Lemma